FACULDADE DE
CIÊNCIAS E TECNOLOGIA
DEPARTAMENTO DE INFORMÁTICA

Universality of Consensus

lectures 13 & 14 (2025-04-28)

Master in Computer Science and Engineering

— Concurrency and Parallelism / 2024-25 —

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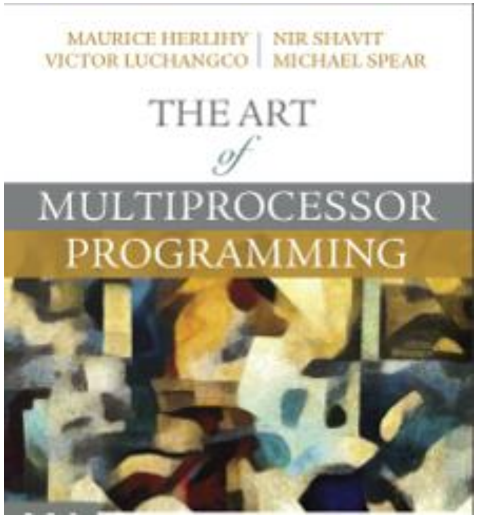
Based in the companion slides from "The Art of Multiprocessor Programming"

Outline

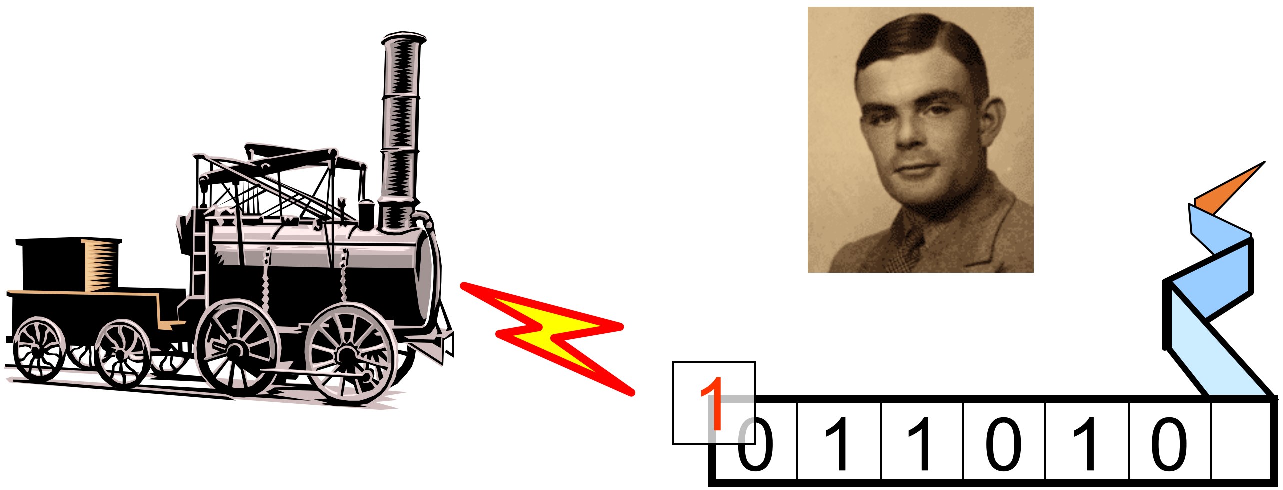
- Consensus
  - Universality
  - A lock-free universal construction
  - A wait-free universal construction



  - Bibliography:
    - **Chapters 6** of book

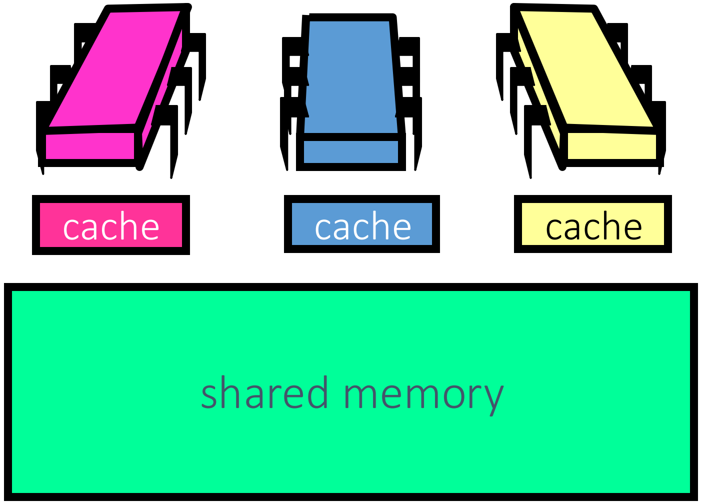
      Herlihy M., Shavit N., Luchangco V., Spear M.;
      The Art of Multiprocessor Programming;
      Morgan Kaufmann (2020); ISBN: 978-0-12-415950-1

Turing Computability



- A mathematical model of computation
- Computable = Computable on a T-Machine

Shared-Memory Computability

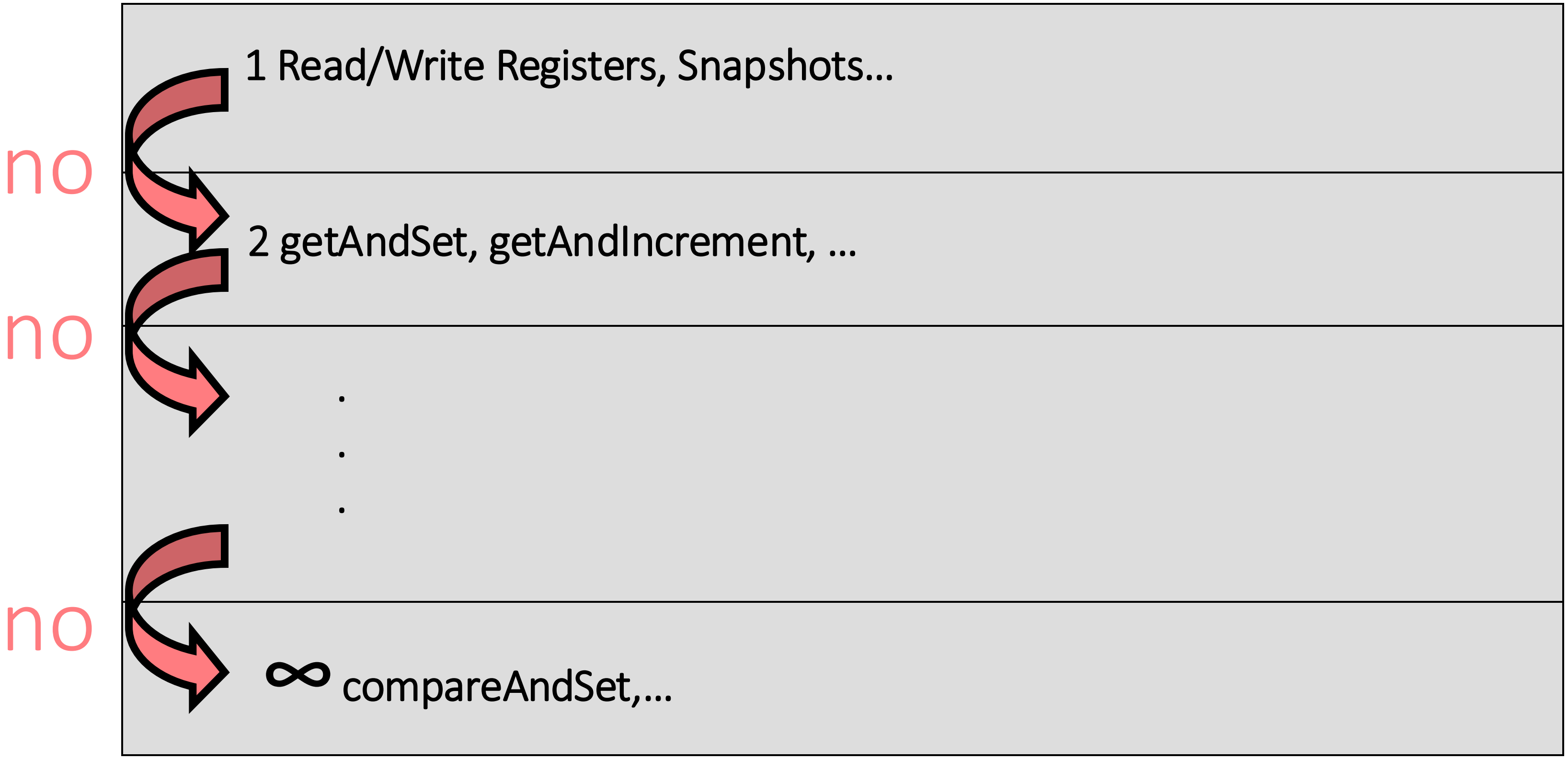


- Model of asynchronous concurrent computation
- Computable = Wait-free/Lock-free computable on a multiprocessor

Consensus Hierarchy

1 Read/Write Registers, Snapshots…
2 getAndSet, getAndIncrement, …
. . .
∞ compareAndSet,…

Who Implements Whom?



Hypothesis


1 Read/Write Registers, Snapshots…
yes?
2 getAndSet, getAndIncrement, …
yes?
.
.
.
yes?
∞ compareAndSet,…

Theorem: Universality

- Consensus is universal
- From *n*-thread consensus build a
  - Wait-free
  - Linearizable
  - ***n***-threaded implementation
  - Of **any** sequentially specified object

Proof Outline

- A universal construction
  - From ***n***-consensus objects
  - And atomic registers
- Any wait-free linearizable object
  - Not a practical construction
  - But we know where to start looking …

Like a Turing Machine

- This construction
  - Illustrates what needs to be done
  - Low-hanging fruit for optimization
- Correctness, not efficiency
  - **Why** does it work? (Asks the scientist)
  - **How** does it work? (Asks the engineer)

A Generic Sequential Object

```
public interface SeqObject {
  public abstract
    Response apply(Invocation invoc);
}
```

A Generic Sequential Object

```
public interface SeqObject {
  public abstract
    Response apply(Invocation invoc);
}
```

Push:5, Pop:null

Invocation

```java
public class Invoc {
 public String method;
 public Object[] args;
}
```

Invocation

```
public class Invoc {
  public String method;
  public Object[] args;
}
```

Method name

Invocation

```
public class Invoc {
 public String method;
 public Object[] args;
}
```

Arguments

A Generic Sequential Object

```
public interface SeqObject {
  public abstract
    Response apply(Invocation invoc);
}
```

A Generic Sequential Object

```
public interface SeqObject {
  public abstract
    Response apply(Invocation invoc);
}
```

OK, 4

Response

```
public class Response {
  public Object value;
}
```

Response

```
public class Response {
   public Object value;
}
```

Return value

Universal Concurrent Object

```
public interface SeqObject {
  public abstract
    Response apply(Invocation invoc);
}
```

A universal concurrent object
is linearizable to the generic
sequential object

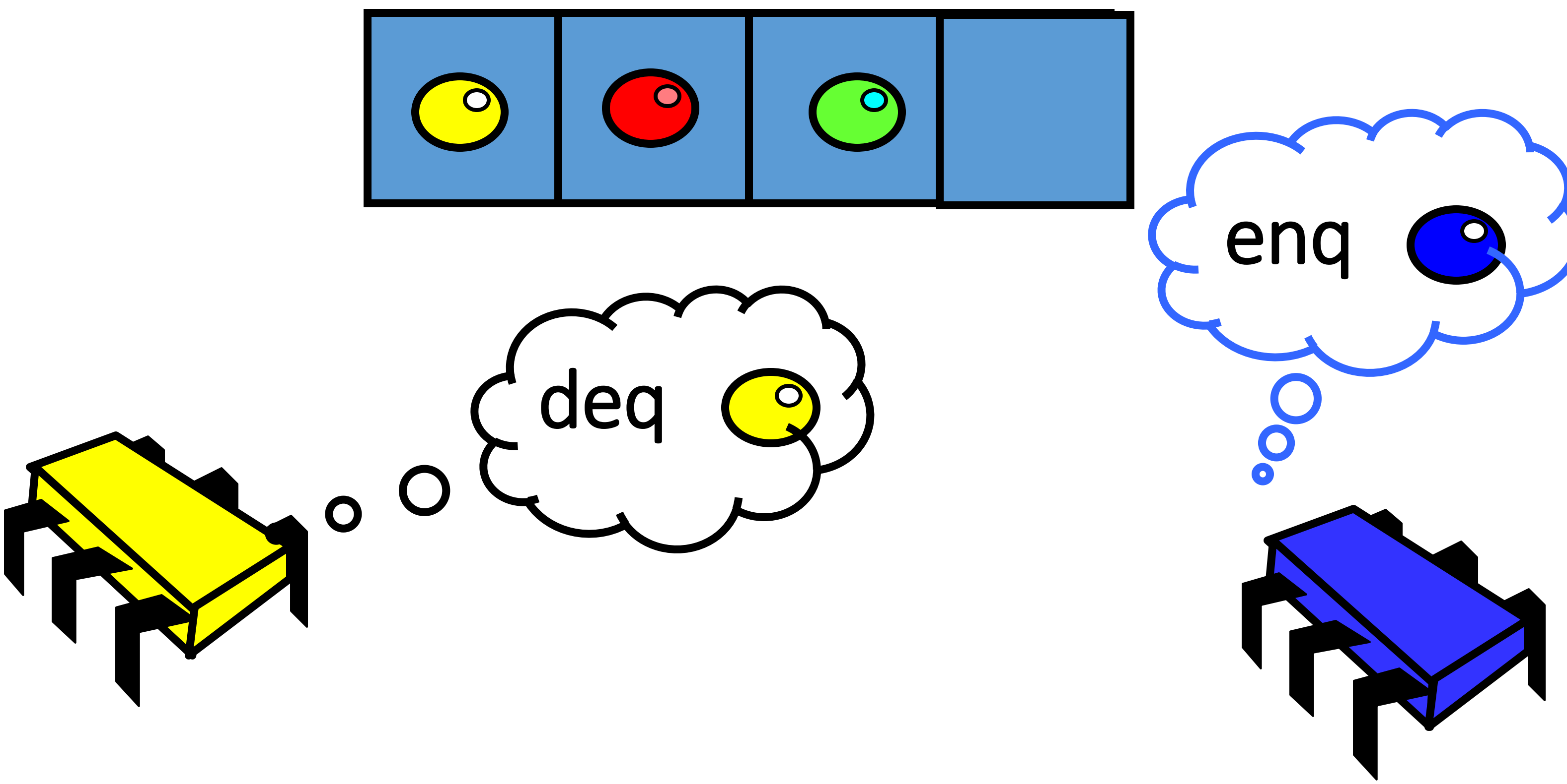
Start with Lock-Free Universal Construction

- First Lock-free: infinitely often some method call finishes.
- Then Wait-Free: each method call takes a finite number of steps to finish.

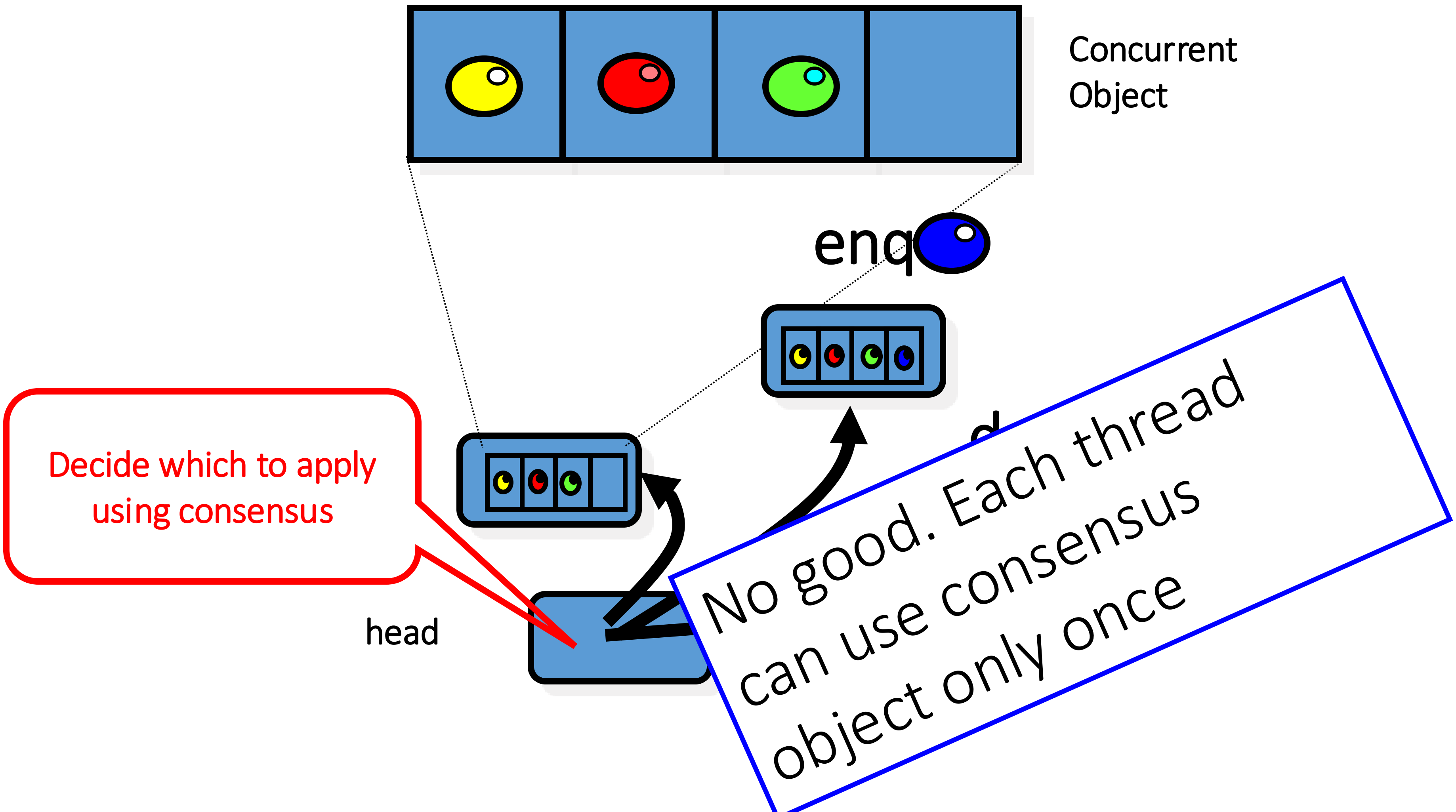
Naïve Idea

- Consensus object stores reference to cell with current state
- Each thread creates new cell
  - computes outcome,
  - tries to switch pointer to its outcome
- Sadly, no …
  - consensus objects can be used once only

Naïve Idea



Naïve Idea



Concurrent Object
enq
Decide which to apply using consensus
head
No good. Each thread can use consensus object only once

Once is not Enough?

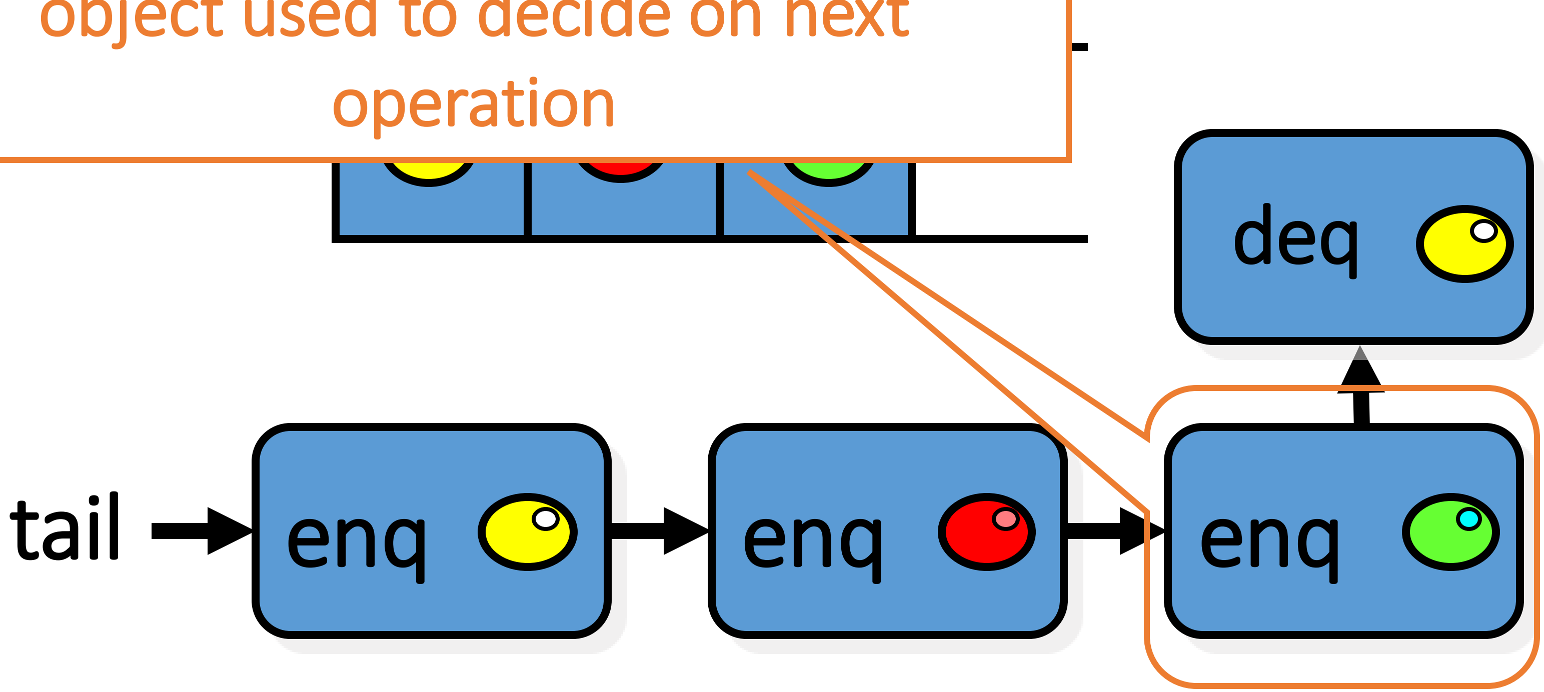
Queue based consensus

```
public T deq(T value) {
  propose(value);
  Ball ball = queue.deq();
  if (ball == Ball.RED)
   return proposed[i];
  else
   return proposed[1-i];
}
```

Solved one-shot 2-consensus.
Not clear how to reuse or reread …

Improved Idea: Linked-List Representation

Each node contains a fresh consensus object used to decide on next operation



Universal Construction

- Object represented as
  - Initial Object State
  - A Log: a linked list of the method calls
- New method call
  - Find end of list
  - Atomically append call
  - Compute response by replaying log

Basic Idea

- Threads use one-time consensus object to decide which node goes next
- Threads update actual next field to reflect consensus outcome
  - OK because they all write the same value
- Challenges
  - Lock-free means we need to worry what happens if a thread stops in the middle

Basic Data Structures

```
public class Node implements
java.lang.Comparable {
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 public Node(Invoc invoc) {
    invoc = invoc;
    decideNext = new Consensus<Node>()
    seq = 0;
  }
```

Basic Data Structures

```
public class Node implements
java.lang.Comparable {
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 pu
    decideNext = new Consensus<Node>();
    seq = 0;
  }
```

Standard interface for class whose objects are totally ordered

Basic Data Structures

```
public class Node implements
java.lang.Comparable {
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 pu
     invoc = invoc;
     decideNext = new Consensus<Node>()
     seq = 0;
   }
```

the invocation

Basic Data Structures

```
public class Node implements
java.lang.Comparable {
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 public Node(Invoc invoc) {
    invoc = invoc;
  }
```

Decide on next node

(next method applied to object)

Basic Data Structures

```
public class Node implements
java.lang.Comparable {
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 public Node(Invoc invoc) {
```

Traversable pointer to next node
(needed because you cannot repeatedly read a consensus object)

```
 }
```

Basic Data Structures

```
public class Node implements
java.lang.Comparable {
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 public Node(Invoc invoc) {
    invoc = invoc;
    decideNext = new Consensus<Node>()
    seq = 0;
  }
```

Seq number

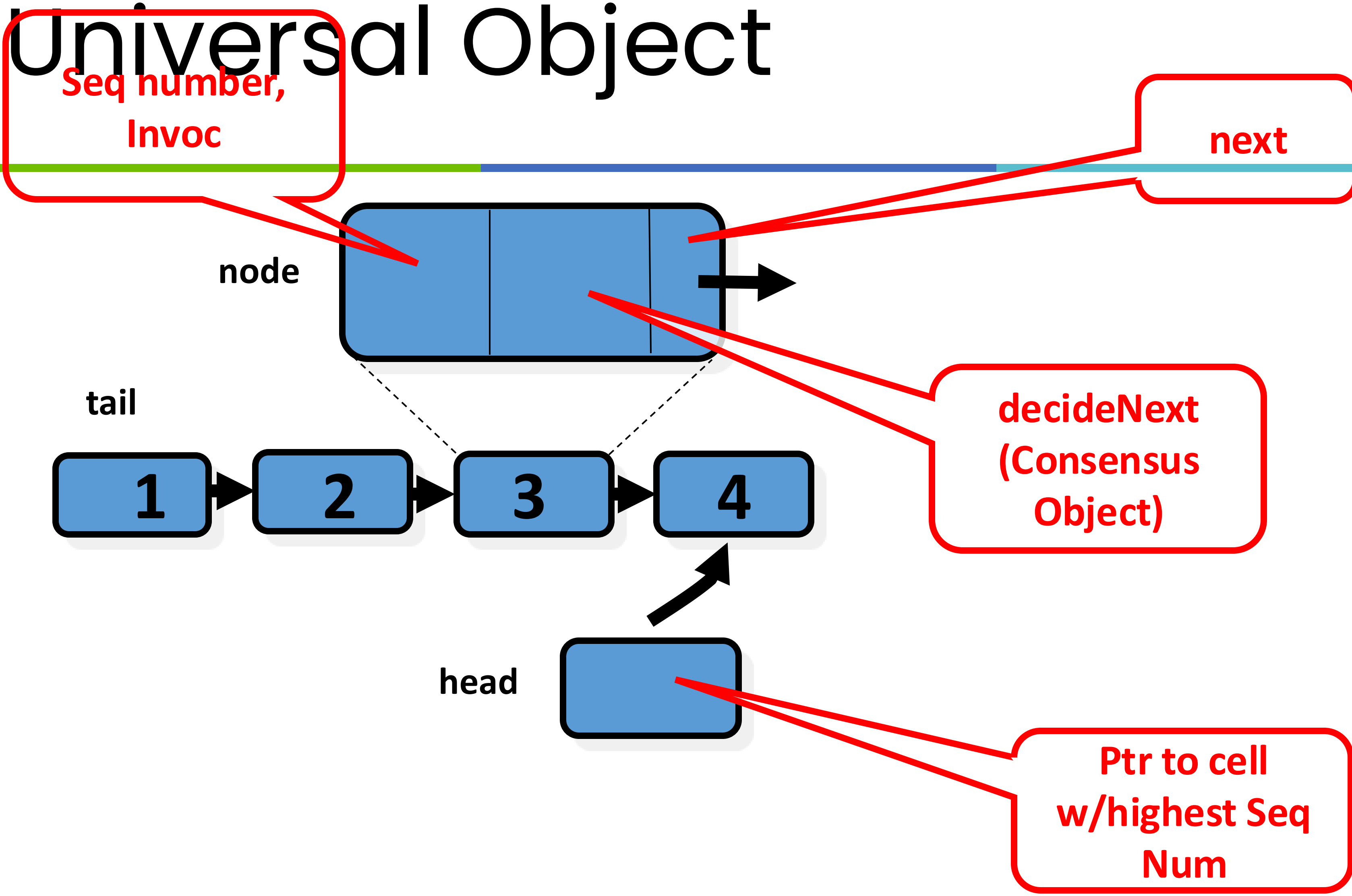
Basic Data Structures

```
public class Node implements
```

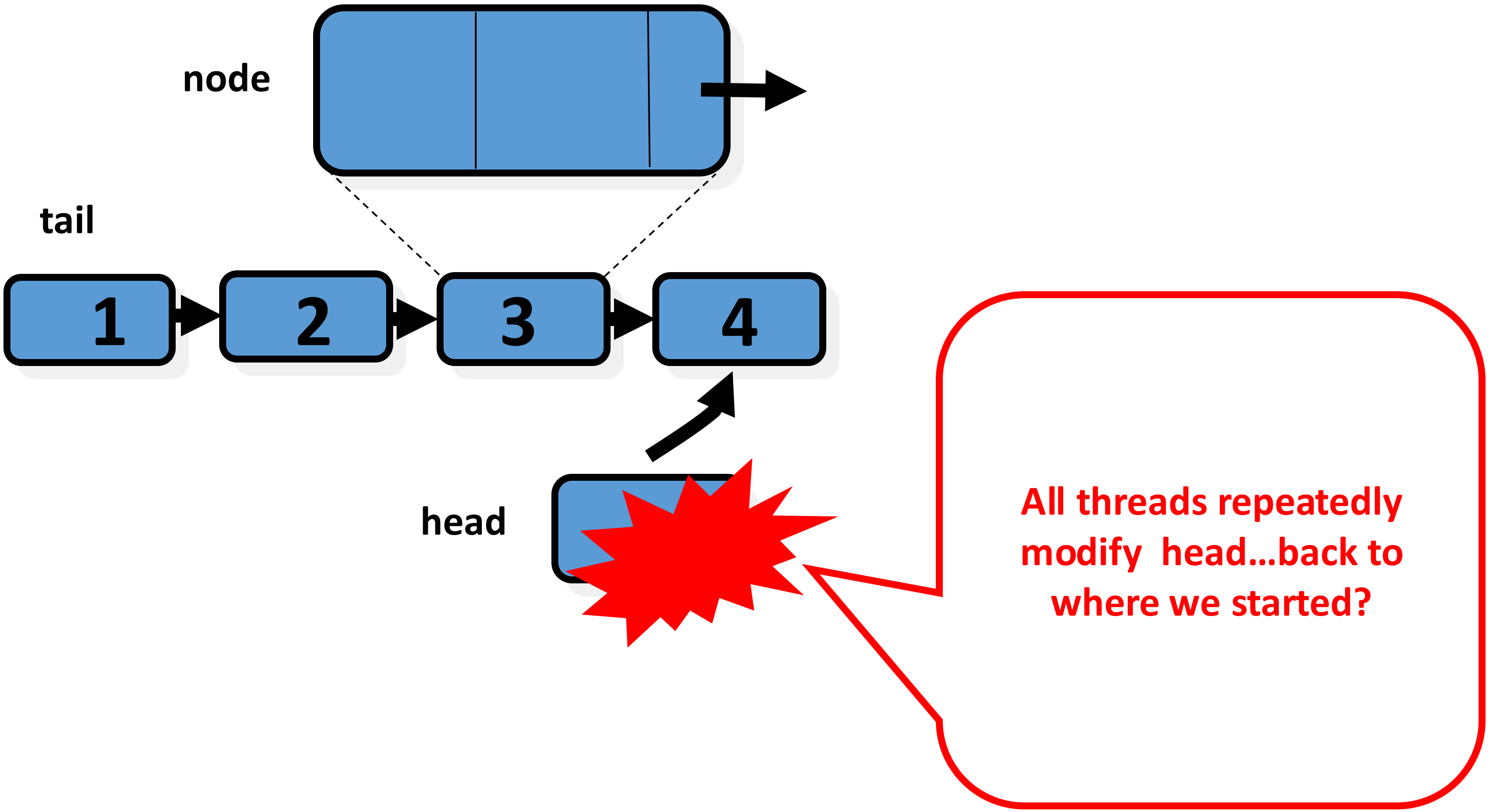
Create new node for this method invocation

```
 public Invoc invoc;
 public Consensus<Node> decideNext;
 public Node next;
 public int seq;
 public Node(Invoc invoc) {
    invoc = invoc;
    decideNext = new Consensus<Node>()
    seq = 0;
  }
```

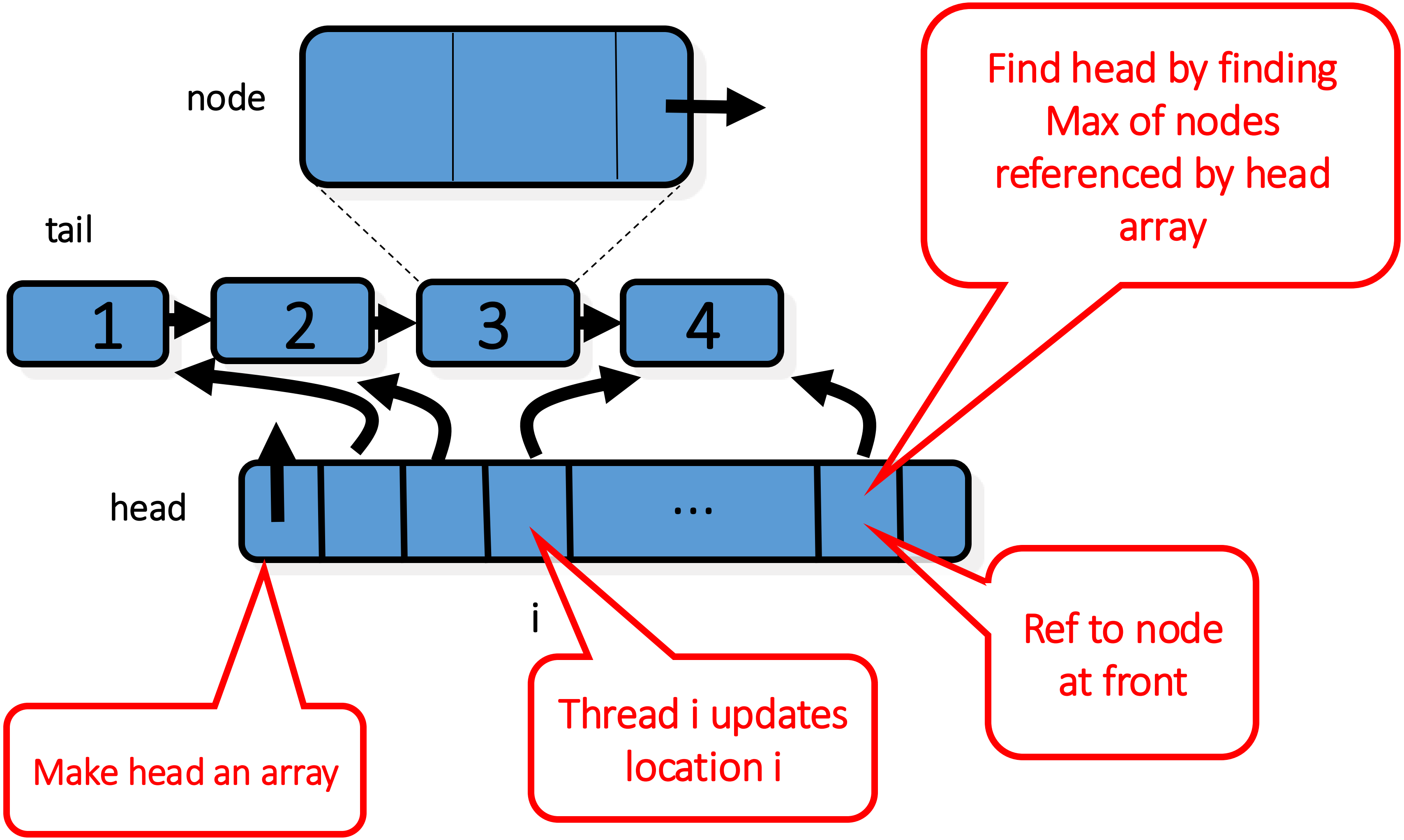
Universal Object



Universal Object



The Solution



Universal Object

```
public class Universal {
  private Node[] head;
  private Node tail = new Node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
        head[j] = tail
   }
```

Universal Object

```
public class Universal {
  private Node[] head;
  private Node tail = new Node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
        head[j] = tail
   }
```

Head Pointers Array

Universal Object

```
public class Universal {
  private Node[] head;
  private Node tail = new Node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
        head[j] = tail
   }
```

Tail is a sentinel node with sequence number 1

Universal Object

```
public class Universal {
  private Node[] head;
  private Node tail = new Node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
        head[j] = tail
  }
```

Initially head points to tail

Find Max Head Value

```
public static Node max(Node[] array) {
    Node max = array[0];
    for (int i = 1; i < array.length; i++)
      if (max.seq < array[i].seq)
        max = array[i];
      return max;
  }
```

Find Max Head Value

```
public static Node max(Node[] array) {
    Node max = array[0];
    for (int i = 1; i < array.length; i++)
      if (max.seq < array[i].seq)
        max = array[i];
      return max;
  }
```

Traverse the array

Find Max Head Value

```
public static Node max(Node[] array) {
    Node max = array[0];
    for (int i = 1; i < array.length; i++)
      if (max.seq < array[i].seq)
        max = array[i];
      return max;
  }
```

Compare the seq nums of nodes pointed to by the array

Find Max Head Value

```
public static Node max(Node[] array) {
    Node max = array[0];
    for (int i = 1; i < array.length; i++)
      if (max.seq < array[i].seq)
        max = array[i];
      return max;
  }
```

Return node with maximal sequence number

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    head[i] = after;
    }
  …
```

Universal Application Part I

```
public Response apply(Invoc invoc) {
   int i = ThreadID.get();
   Node prefer = new node(invoc);
   while (prefer.seq == 0) {
     Node before = Node.max(head);
     Node after =
      before.decideNext.decide(prefer);
     before.next = after;
     after.seq = before.seq + 1;
```

Apply has invocation as input and returns the appropriate response

…

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    he
    }
  …
```

my ID

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    head[i] = after;
    }
  …
```

My method call

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.
    after.s
    head[i] = after;
    }
  …
```

As long as I have not been threaded into list

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.
    after.s
    head[i] = after;
    }
  …
```

Head of list to which we will try to append

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    head[i
    }
  …
```

Propose next node

Universal Application

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
```

Set next field to consensus winner

```
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    head[i] = after;
    }
  …
```

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  Node prefer = new node(invoc);
```

Set seq number
(indicating node was appended)

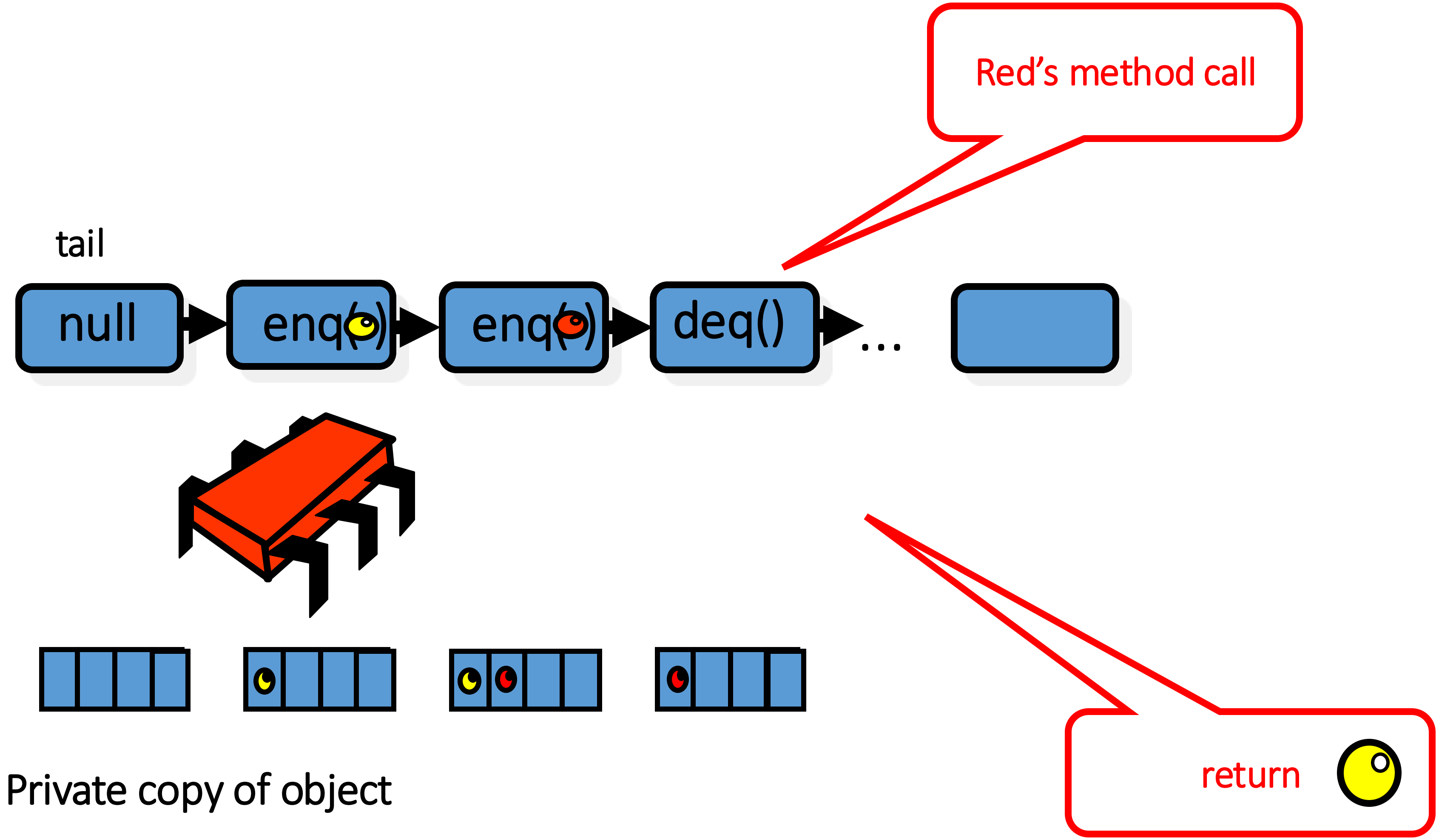
```
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    head[i] = after;
    }
  …
```

Universal Application Part I

```
public Response apply(Invoc invoc) {
  int i =
  Node pre
  while (prefer.seq == 0) {
    Node before = Node.max(head);
    Node after =
     before.decideNext.decide(prefer);
    before.next = after;
    after.seq = before.seq + 1;
    head[i] = after;
   }
  …
```

add to head array so new head will be found

Part 2 – Compute Response



Universal Application Part 2

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

Universal Application Part II

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
   }
```

Compute result by sequentially applying method calls in list to a private copy

Universal Application Part II

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

Start with copy of sequential object

Universal Application Part II

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

new method call appended after tail

Universal Application Part II

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

While my method call not linked …

Universal Application Part II

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
  MyObject.apply(current.invoc);
  current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

Apply current node's method

Universal Application Part II

```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

Return result after my method call applied

Correctness

- List defines linearized sequential history
- Thread returns its response based on list order

Lock-freedom

- Lock-free because
  - A thread moves forward in list
  - Can repeatedly fail to win consensus on “real” head only if another succeeds
  - Consensus winner adds node and completes within a finite number of steps

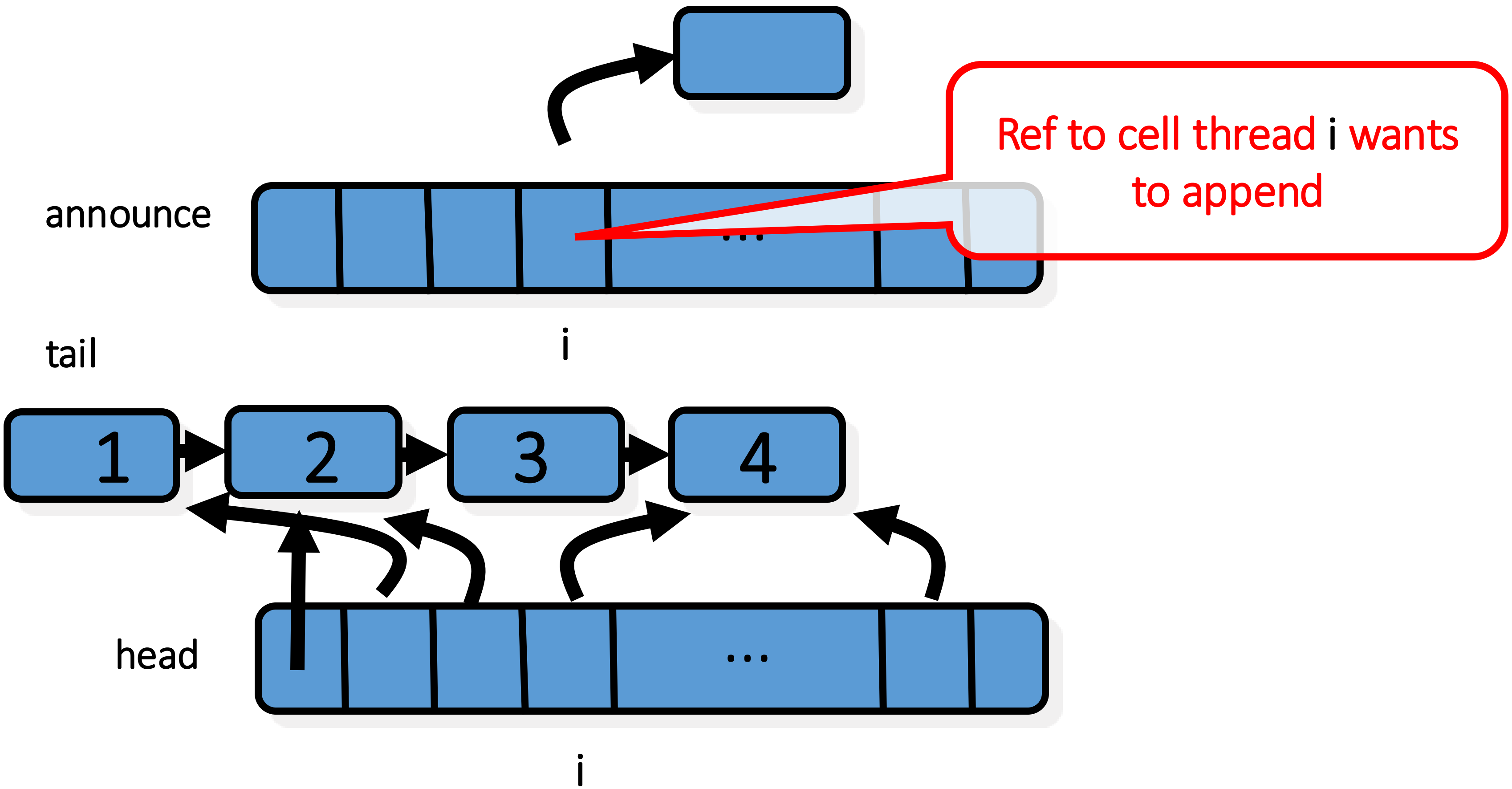
Wait-free Construction

- Lock-free construction + `announce` array
- Stores (pointer to) node in `announce`
  - If a thread doesn't append its node
  - Another thread will see it in array and ***help*** append it

Helping

- "Announcing" my intention
  - Guarantees progress
  - Even if the scheduler hates me
  - My method call will complete
- Makes protocol wait-free
- Otherwise starvation possible

Wait-free Construction



The Announce Array

```
public class Universal {
  private Node[] announce;
  private Node[] head;
  private Node tail = new node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
    head[j] = tail; announce[j] = tail;
  }
```

The Announce Array

```
public class Universal {
  private Node[] announce;
  private Node[] head;
  private Node tail = new node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
    head[j] = tail; announce[j] = tail;
  }
```

New field: announce array

The Announce Array

```
public class Universal {
  private Node[] announce;
  private Node[] head;
  private Node tail = new node();
  tail.seq = 1;
  for (int j=0; j < n; j++){
    head[j] = tail; announce[j] = tail;
  }
```

All entries initially point to tail

A Cry For Help

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  announce[i] = new Node(invoc);
  head[i] = Node.max(head);
  while (announce[i].seq == 0) {
  …
  // while node not appended to list
  …
  }
```

A Cry For Help

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  announce[i] = new Node(invoc);
  head[i] = Node.max(head);
  while (announce[i].seq == 0) {
  …
  // while node not appended to list
  …
  }
```

Announce new method call, asking help from others

A Cry For Help

```
public Response apply(Invoc invoc) {
  int i = ThreadID.get();
  announce[i] = new Node(invoc);
  head[i] = Node.max(head);
  while (announce[i].seq == 0) {
  …
  // while node not appended to list
  …
  }
```

Look for end of list

A Cry For Help

```
public Response apply(Invoc invoc) {
   int i = ThreadID.get();
   announce[i] = new Node(invoc);
   head[i] = Node.max(head);
   while (announce[i].seq == 0) {
   …
   // while node not appended to list
   …
   }
```

Main loop, while node not appended (either by me or helper)

Main Loop

- Non-zero sequence # means success
- Thread keeps helping append nodes
- Until its own node is appended

Main Loop

```
while (announce[i].seq == 0) {
 Node before = head[i];
 Node help = announce[(before.seq + 1) % n];
 if (help.seq == 0)
     prefer = help;
    else
     prefer = announce[i];
…
```

Main Loop

```
while (announce[i].seq == 0) {
 Node before = head[i];
 Node help = announce[(before.seq + 1) % n];
 if (help.seq == 0)
     prefer = help;
   else
```

Keep trying until my cell gets a sequence number

Main Loop

```
while (announce[i].seq == 0) {
 Node before = head[i];
 Node help = announce[(before.seq + 1) % n];
 if (help.seq == 0)
     prefer = help;
    else
     prefer = announce[i];
```

Possible end of list

Main Loop

```
while (announce[i].seq == 0) {
 Node before = head[i];
 Node help = announce[(before.seq + 1) % n];
 if (help.seq == 0)
     prefer = help;
    else
     prefer = announce[i];
```

Whom do I help?

Altruism

- Choose a thread to “help”
- If that thread needs help
  - Try to append its node
  - Otherwise append your own
- Worst case
  - Everyone tries to help same pitiful loser
  - Someone succeeds

Help!

- When last node in list has sequence number *k*
- All threads check …
  - –Whether thread k+1 mod n wants help
  - –If so, try to append her node first

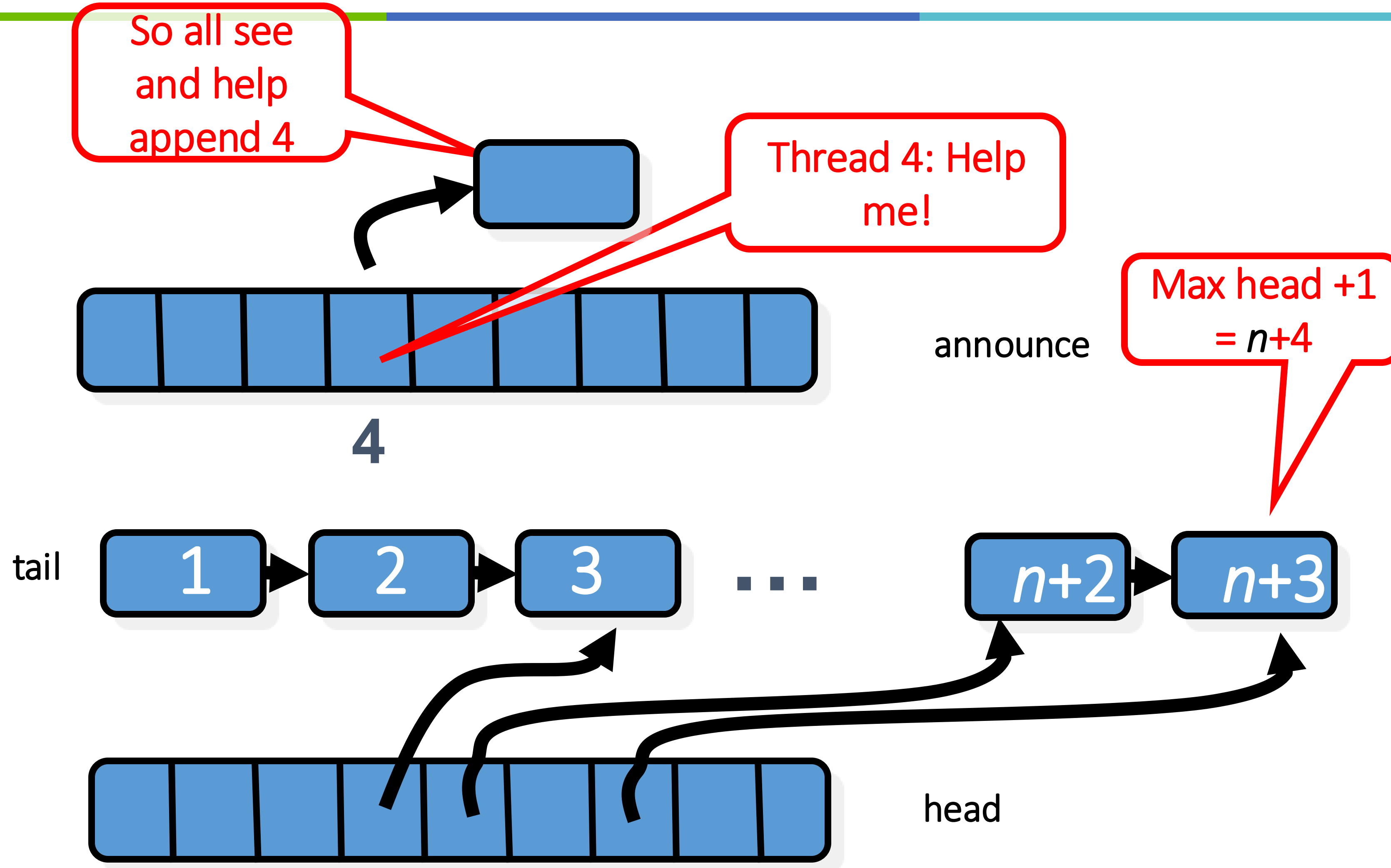
Help!

- First time after thread k+1 announces
  - No guarantees
- After n more nodes appended
  - Everyone sees that thread k+1 wants help
  - Everyone tries to append that node
  - Someone succeeds

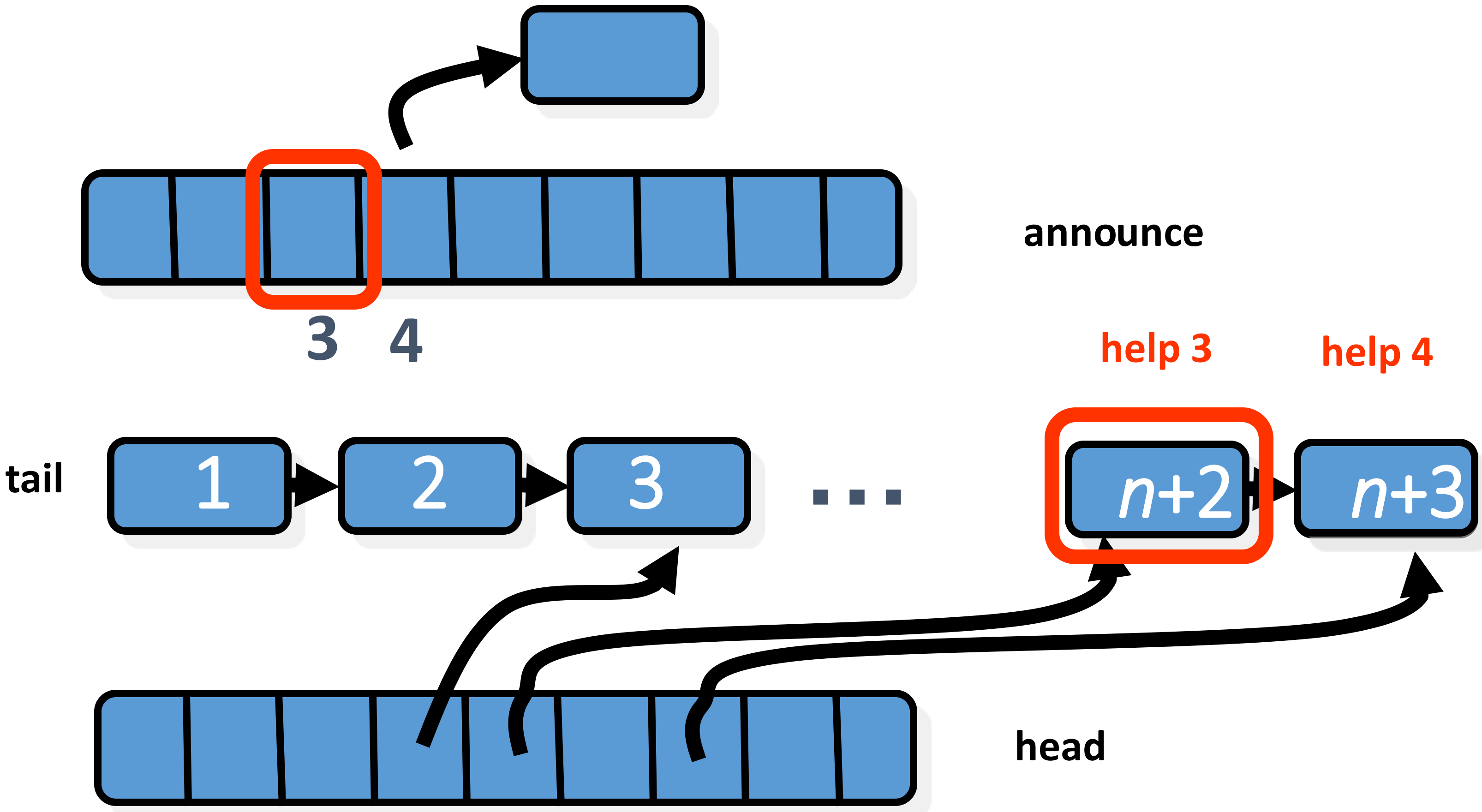
Sliding Window Lemma

- After thread A announces its node
- No more than *n* other calls
  - Can start and finish
  - Without appending A's node

Helping


So all see and help append 4
Thread 4: Help me!
Max head +1 = n+4
announce
4
tail
1
2
3
...
n+2
n+3
head

The Sliding Help Window



Sliding Help Window

```
while (announce[i].seq == 0) {
 Node before = head[i];
 Node help = announce[(before.seq + 1) % n];
 if (help.seq == 0)
     prefer = help;
    else
     prefer = announce[i];
```

In each main loop iteration pick another thread to help

Sliding Help Window

Help if help required, but otherwise it's all about me!

```
while (ann
 Node before = head[i];
 Node help = announce[(before.seq + 1) % n];
 if (help.seq == 0)
     prefer = help;
    else
     prefer = announce[i];
…
```

Rest is Same as Lock-free

```
while (announce[i].seq == 0) {
   …
   Node after =
    before.decideNext.decide(prefer);
   before.next = after;
   after.seq = before.seq + 1;
   head[i] = after;
  }
```

Rest is Same as Lock-free

```
while (announce[i].seq == 0) {
  …
  Node after =
   before.decideNext.decide(prefer);
  before.next = after;
  after.seq = before.seq + 1;
  head[i] = after;
 }
```

Call consensus to attempt to append

Rest is Same as Lock-free

```
while (announce[i].seq == 0) {
  Node after =
   before.decideNext.decide(prefer);
  before.next = after;
  after.seq = before.seq + 1;
  head[i] = after;
 }
```

cache consensus result for later use

Rest is Same as Lock-free

```
while (announce[i].seq == 0) {
  …
   before.decideNext.decide(prefer);
  before.next = after;
  after.seq = before.seq + 1;
  head[i] = after;
 }
```

Tell world that node is appended

Finishing the Job

- Once thread's node is linked …
- The rest same as lock-free algorithm
- Compute result by
  - sequentially applying list's method calls
  - to a private copy of the object
  - starting from the initial state

Then Same Part II

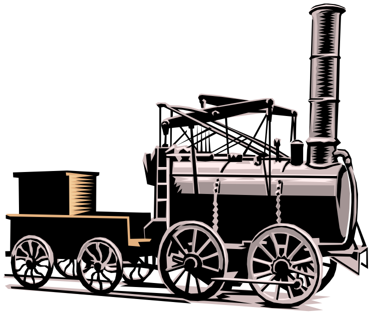
```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
while (current != prefer){
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

Universal Application Part II

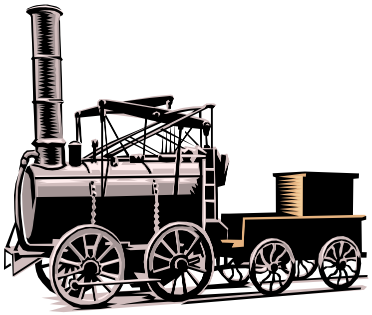
```
...
//compute my response
SeqObject MyObject = new SeqObject();
current = tail.next;
   MyObject.apply(current.invoc);
   current = current.next;
  }
return MyObject.apply(current.invoc);
}
```

Return result after applying my method

Shared-Memory Computability



Universal Construction



Wait-free/Lock-free computable

=

Solving n-consensus

Swap (getAndSet) not Universal

```
public class RMWRegister {
 private int value;
 public boolean getAndSet(int update) {
  int prior = value;
  value = update;
  return prior;
 }
}
```

Swap (getAndSet) not Universal

```
public class RMWRegister {
 private int value;
 public boolean getAndSet(int update) {
  int prior = value;
  value = update;
  return prior;
 }
}
```

Consensus number 2

Swap (getAndSet) not Universal

```
public class RMWRegister {
 private int value;
 public boolean getAndSet(int update) {
  int prior = value;
  value = update;
  return prior;
 }
}
```

Not universal for ≥3 threads

CompareAndSet is Universal

```
public class RMWRegister {
 private int value;
 public boolean
   compareAndSet(int expected,
                 int update) {
  int prior = value;
  if (value == expected) {
   value = update;
   return true;
  }
 return false;
 }}
```

CompareAndSet is Universal

```
public class RMWRegister {
 private int value;
 public boolean
   compareAndSet(int expected,
                 int update) {
   int prior = value;
   if (value == expected) {
    value = update;
    return true;
   }
 return false
 }}
```

Consensus number ∞

CompareAndSet is Universal

```
public class RMWRegister {
 private int value;
 public boolean
   compareAndSet(int expected,
                 int update) {
   int prior = value;
   if (value == expected) {
    value = update;
    return true;
   }
```

Universal for any number of threads

On Older Architectures

- IBM 360
  - testAndSet (getAndSet)
- NYU UltraComputer
  - getAndAdd (fetchAndAdd)
- Neither universal
  - Except for 2 threads

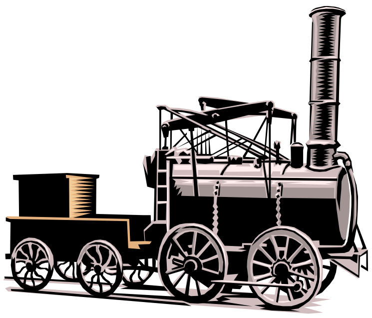
On Newer Architectures

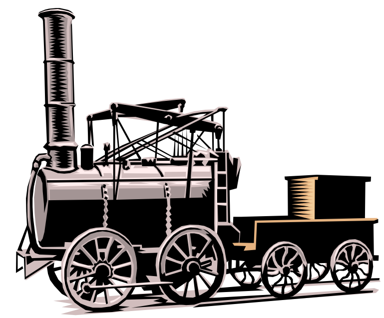
- Intel x86, Itanium, SPARC
  - compareAndSet (CAS, CMPXCHG)
- Alpha AXP, PowerPC
  - Load-locked/store-conditional
- All universal
  - For any number of threads
- Trend is clear…

Practical Implications

- Any architecture that does not provide a universal primitive has inherent limitations
- You cannot avoid locking for concurrent data structures …

Shared-Memory Computability





Universal Construction

Wait-free/Lock-free computable

=

Solving *n*-consensus

Veni, Vidi, Vici

- We saw
  - how to define concurrent objects
- We discussed
  - computational power of machine instructions
- Next
  - use these foundations to understand the real world

The END